

BRILLIANT STUDENTS FORM 1 END TERM 2
SERIES 3 EXAMS (ALL SUBJECTS TESTED)



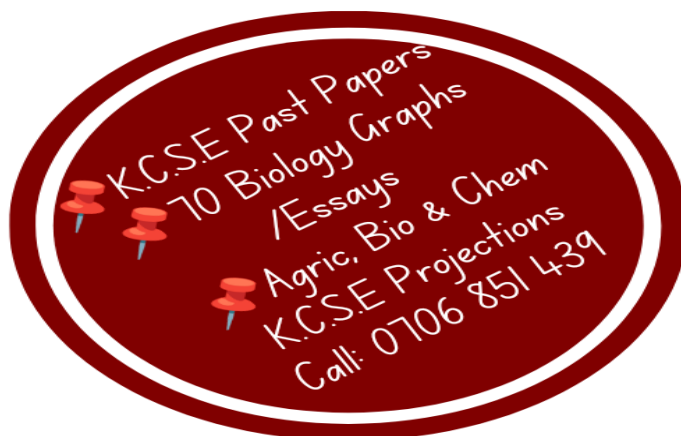
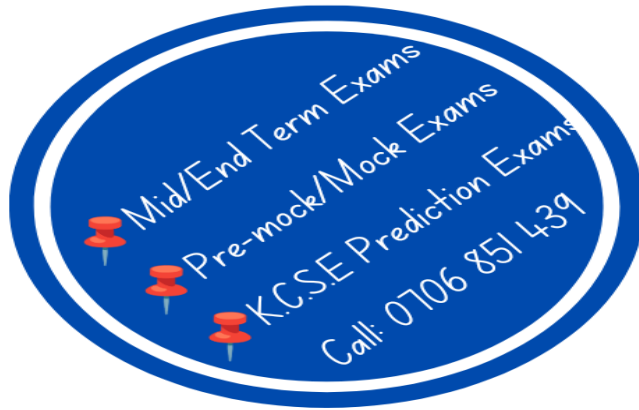
BRILLIANT STUDENTS FORM 1
END TERM 2 SERIES 3 EXAMS

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BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

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FORM 1
AGRICULTURE
SECTION A (40MKS)

1.Name three conditions under which shifting cultivation is practicable. **(3mks)**
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2.List four disadvantages associated with the burning of land in shifting land preparation. **(4mks)**
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3.Outline four environmental conditions that may lead to poor crop yields. **(4mks)**
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4.Name four human factors that influence production and distribution of crops and livestock **(4mks)**

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5a) What is sub –soiling? **(1mk)**

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b) Give four advantages of minimum tillage. **(4mks)**

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6. Mention three types of pumps. **(3mks)**

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7. Outline four properties of clean and safe water. **(4mks)**

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8. Name three methods of surface irrigation. (3mks)

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9a) Define drainage. (1mk)

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b) Give three causes of poor drainage. (3mks)

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10. There are several agricultural practices that causes water pollution. Briefly mention them (4mks)

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11. Name a tool used for each of the following operations. (2mks)

i)Testing mastitis

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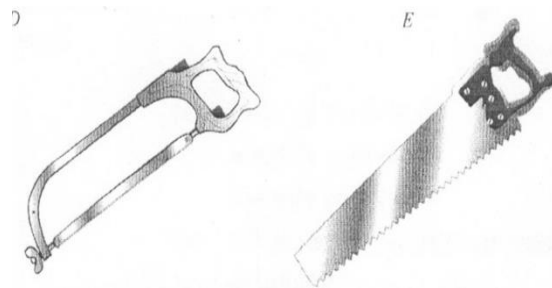


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ii) Cut overgrown hooves
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SECTION B (30MKS)

Answer all the questions in this section.

12. Diagram D and E below are illustrations of workshop tools. Study them and answer the questions that follow.



a) Identify the tools marked D and E. (2mks)

D

E

b) State one function of each of the tool shown. (2mks)

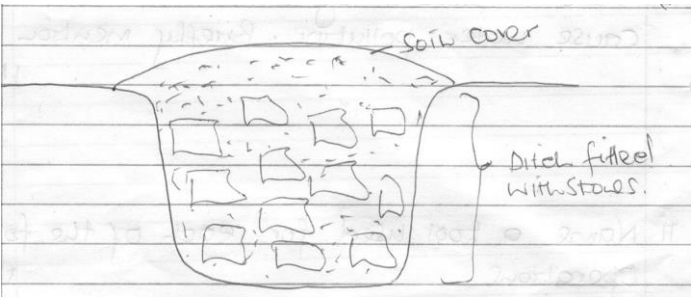
D
.....
.....
E
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c) Give one maintenance practice of tool E. (1mk)

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13a) Identify the method of drainage shown below.

(1mk)



b) Other than the method shown in (a) above list other four methods used in draining farm land. (2mks)

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C) Outline four reasons for draining farm land.

(2mks)

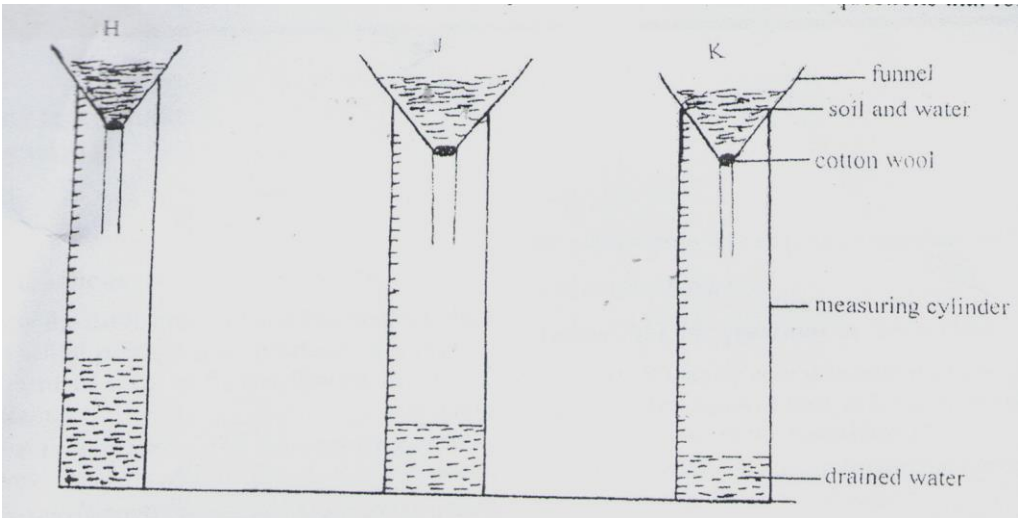
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14. The illustrations below represent an experiment to compare the porosity and water holding capacity of three of soils. Carefully study the experiment and then answer the questions that follow.



a i) Identify the soil in each of the funnel labeled. (3mks)

H.

J.

K.

b) Which of the three types of soil can be said to have the highest porosity rate? (1mk)

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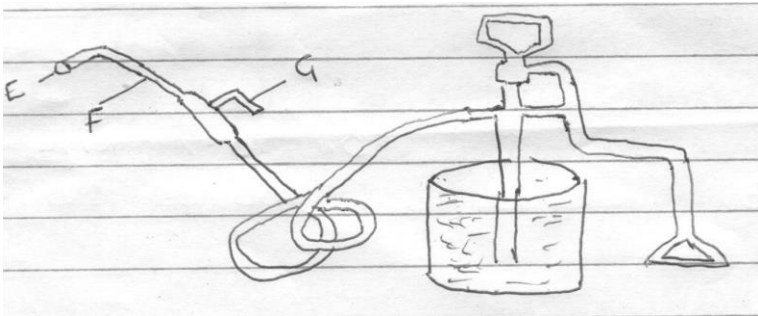
C)Which type of soil would be suitable for planting paddy rice? (1mk)

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15. Complete the table below that attempts to classify soil based on the sizes of soil particles. (5mks)

Particles	Size (diameter) in MM
	Above 2.0mm
	Between 0.20mm and 2.00mm
	Between 0.20mm and 0.20mm
	Between 0.002mm and 0.02mm
	Below 0.002mm

16. Below is a diagram of farm equipment. Use it to answer the questions that follows.



a) Identify the equipment. **(1mk)**

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.....

b) State the use of the equipment. **(1mk)**

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c) Name the parts labeled E, F and G. **(3mks)**

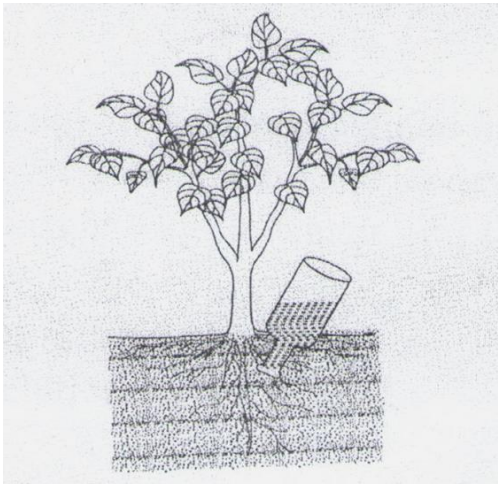
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17a) Identify the method of irrigation shown below. **(1mk)**



b) What are the disadvantages of this method of irrigation? **(2mks)**

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c)Outline how the method of irrigation in (a) above is maintained? (2mks)

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SECTION C(30MKS)

Answer all question in this section.

18a) Outline six biotic factors influencing agriculture. (6mks)

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b) State the four aspects of rainfall that are important for crop growth . (4mks)

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19a) State five factors that determine the number of times secondary cultivation is done.(5mks)

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b) Outline the advantages of minim tillage.

(5mks)

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20a) State four importance of water treatment.

(4mks)

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b) What factors determine the choice of the type of irrigation used?

(6mks)

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

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FORM 1

BIOLOGY

END TERM 2

Answer all the questions in the spaces provided(100mks)

1. a.Name one product of the light stage during photosynthesis **(1mk)**

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b.Name the process by which water molecules are split by light during photosynthesis **(1mk)**

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2. What name is given to the process whereby :

a.Red blood cells wrinkle after losing their water to a hypertonic solution? **(1mk)**

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.....

b.Plant cells become flaccid after losing their water to a hypertonic solution? **(1mk)**

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3. Four equal strips A,B,C and D were cut from a potato whose cells had a sugar concentration of 28.5%.The strips were placed in sugar solutions of different concentrations as follows:

A:10% B 15% C 25% D 35%

i. What changes would you expect in strips A and D? **2mks**

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ii. Account for the change in strip D **(3mks)**

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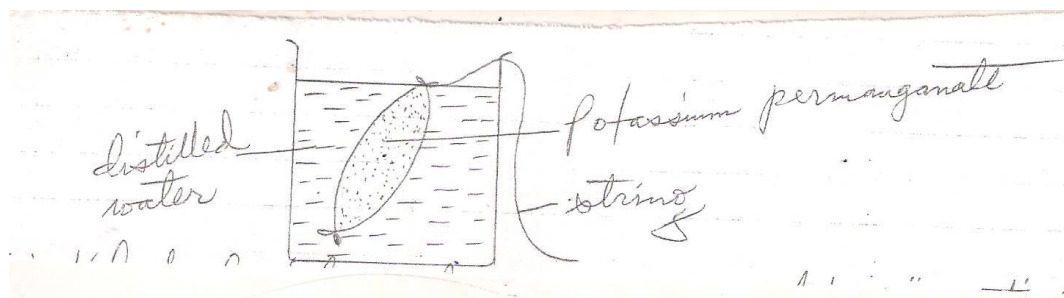
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4. The figure below shows a small piece of visking tubing which is filled with potassium permanganate solution. Its free ends were tied tightly to prevent leakage .It was then dipped in a beaker full of distilled water. The set up was left for 2 hours .Its was observed that the distilled water was coloured purple.



a. What physiological process was being investigated. **(1mk)**

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b. Account for the observation made in (a) above. **(3mks)**

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5. a.State three factors that affect enzyme activities. (3mks)

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b.Name the structures on an enzyme where substrate molecules fix themselves during an enzyme reaction. (1mk)

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c.State what would happen to an enzyme molecule if the temperature is:-

i. Raised above 40°C (1mk)

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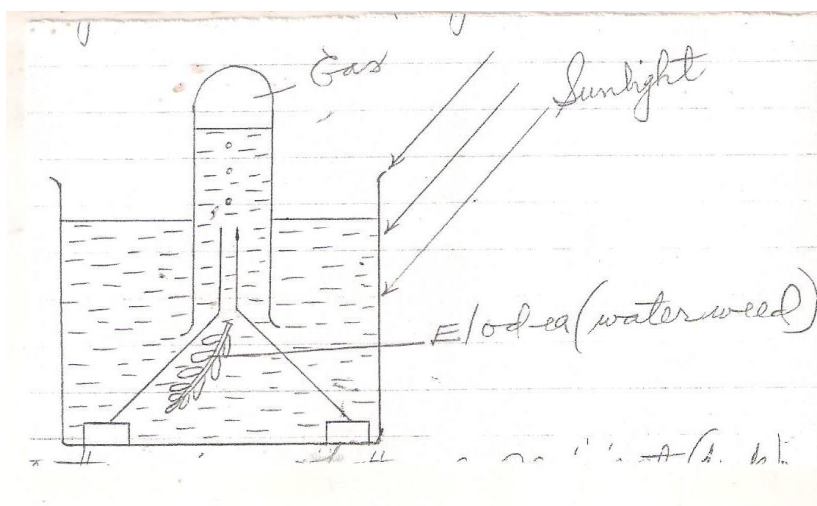
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ii.Lowered below 10°C. (1mk)

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6. The diagram below represents a set up that was used to investigate a certain process in a plant.



a.State the aim of the experiment. **(1mk)**

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b.State a factor that would affect the process. **(1mk)**

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7. State the importance of nucleic acids to an organisms. **(1mk)**

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8. State the significance of the following to a leaf:-

a.Thinness **(1mk)**

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b.Presence of air spaces **(1mk)**

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c.Stomata **(1mk)**

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9. What is the role of roof hairs in plants. **(1mk)**

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10. What is meant by the term organ systems in organisms? **(1mk)**

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11. State three factors that affect the rate of diffusion **(3mks)**

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12. What is compensation point. **(1mk)**

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13. a.State two properties of monosaccharide **.(2mks)**

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b.What is the main function of monosaccharide in organisms? **(1mk)**

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14. State the formula for calculating magnification when using the following

a.Hand lens/naked eyes **(1mk)**

b. A light microscopic **(1mk)**

15. Explain why a mule, a product of mating between a horse and a donkey is sterile. **(1mk)**

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16. State two factors considered while grouping the organisms in the same species. **(2mks)**

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17. Describe how you would carry out food tests to test the presence of starch and reducing sugars on a food sample.

a. Test for starch. **(3mks)**

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b. Test for reducing sugar **(4mks)**

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18. Explain how surface area to volume ratio idea may be applied to explain rate of diffusion in organisms.(2mks)

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19. Which organelles in a cell perform the following functions

a.Manufacture of ribosomes (1mk)

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b.Package cell secretions (1mk)

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c.Energy production (1mk)

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d.Synthesis of carbohydrates. (1mk)

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20. Name the taxonomic unit with:

a.The greatest number of organism. (1mk)

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b.The least number of organisms (1mk)

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21. a. Define the term cell(1mk)

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b. When onion epidermal cells were placed and focused along the diameter of the field of view ,10 cells were viewed and counted. Calculate the length of each epidermal cell in micrometers (1mm = 1000micrometers). Assume the diameter of field of view= 3mm **(2mks)**

22. Name the kingdom in which the organisms named below belong.

a. Bacteria (1mk)

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b. Paramecium (1mk)

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c. Yeast (1mk)

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23. State the deficiency disease caused by deficiency of the following vitamins

a. Vitamins A(1mk)

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b. Vitamin D (1mk)

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c. Vitamin B₁ (1mk)

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24. a. State and explain how the ileum is adapted to perform its function. **(4mks)**

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b.State two functions of the colon during digestion.**(2mks)**

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25. a.Name two diseases that affect human teeth.**(2mks)**

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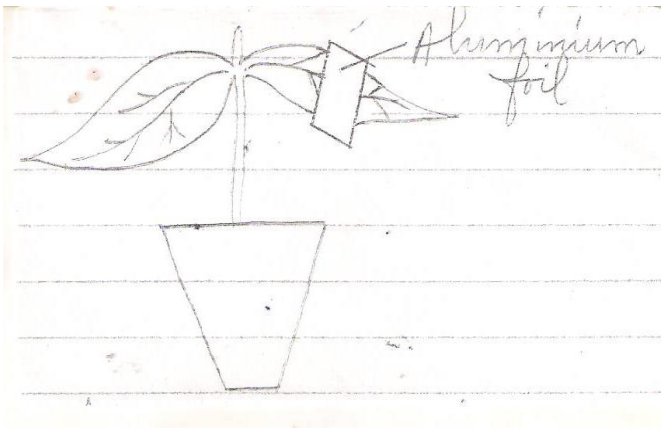
b.(i) Write the dental formula of an adult human. **(1mk)**

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ii)Work out the total number of teeth from the formula above. **(1mk)**

26. In an experiment to investigate a factor affecting photosynthesis, a leaf of a potted plant which had been kept in the dark overnight was covered with aluminium foil as shown in the diagrams below.



The set up was kept in sunlight for three hours after which a food test was carried out on the leaf.

a.Which factor was being investigated in the experiment? (1mk)

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b. Which food test was carried out? (1mk)

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c.i)State the results of the food test. (2mks)

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ii) Account for the results in (c) (i) above.2mks

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d. Why was is necessary to keep the plant in darkness before the experiment? **1mk**

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27. a. Distinguish between the term s homodont and heterodont **(2mks)**

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b.Name the gap between incisors and premolars in some herbivores and state its function. **(2mks)**

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28. State and explain how palisade cells are adapted to perform their function. **(3mks)**

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29. a.Define the term active transport . **(2mks)**

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b.State four factors that affect active transpiration.(4mks)

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30. State the function of the following parts of a light microscope.

a. Coarse adjustment knob. (1mk)

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b.Diaphragm (1mk)

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c. Condenser (1mk)

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31. Name the first four taxa in taxonomy starting from the highest level

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BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

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565/1

BUSINESS STUDIES

INSTRUCTIONS

- (a) Answer all the questions.
- (b) All answers must be written in the spaces provided.
- (c) Answers should be written in english.

1.Outline four purposes of a Business. (4mks)

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2.State four types of business activities. (4mks)

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3.List four disciplines of Business Studies. (4mks)

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4.Outline **four** benefits that Abdi, a student would get from studying Business. **(4mks)**

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5.Below are some of the factors that influence the operations of a business. Indicate with a tick () in the appropriate column, the business environment associated with each of the factors. **(4mks)**

<u>S.no</u>	<u>Factors</u>	<u>Internal</u>	<u>External</u>
	Competition		
	Finance		
	Suppliers		
	N.E.M.A		
	Goals and objectives of the firm.		
	Currency exchange rates.		
	Human resources management		
	Firms organizational structure.		

6.Outline **four** favourable internal factors for business to thrive. **(4mks)**

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7. Give and explain **two** classifications of human wants. (4mks)

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8. Outline **four** characteristics of human wants. (4mks)

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9. Give **four** reasons why Omondi satisfy hunger for food before going to Chaka Ranch for entertainment. (4mks)

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10. Highlight **four** differences between goods and services (4mks)

	<u>Goods</u>	<u>Services</u>
(i)		
(ii)		
(iii)		
(iv)		
(v)		

11. Name **three** classifications of Economic Resources (3mks)

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12. Outline **five** characteristics of economic resources. (5mks)

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13. Indicate whether the following resources are **renewable or non-renewable** (4mks)

<u>Resources</u>	<u>Classification</u>
i. Limestone	
ii. Natural rubber	
iii. Solar energy	
iv. Rivers	
v.Wool	
vi. Cars	
vii. Skin/hide	
viii. Quarry	

14. State the **type of utility** created by each of the following activities (4mks)

<u>ACTIVITY</u>	<u>UTILITY</u>
1.Baking	
2.Banking	
3.Bookselling	
4.Carpentry	
5.Manufacturing	
6.Retailing	
7.Transportation	
8.Tanning leather	

15. Outline **four** disadvantages of Direct Production.(4mks)

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16. Classify each of the following activities in its appropriate **level of production**.(4mks)

<u>Activity</u>	<u>level of production</u>
I. Lumbering	
II. Brick making	
III. Nursing	
IV. Road construction	
V. Legal services	
VI. Quarrying	
VII. Entertainment	
VIII. Chemical making	

17. State **four** features of land as a factor of production.(4mks)

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18. Outline **four** demerits of division of labour. (4mks)

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19. Using appropriate examples, explain the following **types of goods** (4mks)

(i) Producer goods

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(ii) Intermediate goods

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(iii) Economic goods

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(iv) Public goods

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20. Highlight **four** importances of Entrepreneurship in an Economy.(4mks)

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21. List **four** characteristics of an entrepreneur. (4mks)

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22. Outline **four** sources of business ideas. (4mks)

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23. Highlight **four** factors to consider when evaluating a business opportunity (4mks)

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24. List **four** importances of a Business plan.(4mks)

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25. Outline **four** functions of an office.(4mks)

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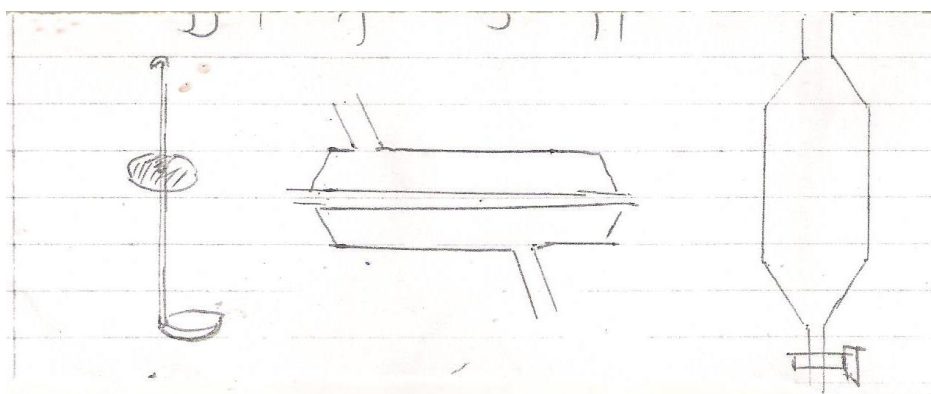
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FORM 1

CHEMISTRY

END TERM 2

1. Identify the following apparatus and their uses. (6mks)



A

B

C

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2.State four preventive measure of drug –abuse. **(4mks)**

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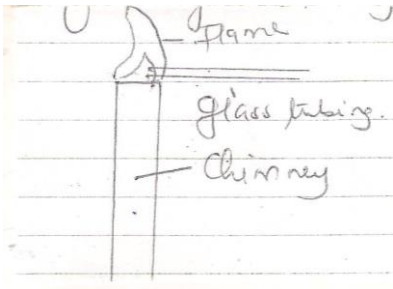
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3.a.State under which conditions are the following flame formed in the laboratory. **2mks**

- i. Luminous.....
- ii. Non- luminous.....

b.In an experiment, a form one student at Kiranja Secondary School placed an end of narrow glass tubing in the inner core of non-luminous flame and lit at the tip of the glass tubing as shown below .



State and explain the observation made at the tip of the glass tubing. **(2mks)**

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4.State one application for each of the following methods of separating mixtures.

a.Filtration **(1mk)**

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b. Fractional distillation (1mk)

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c. Solvent extraction. (1mk)

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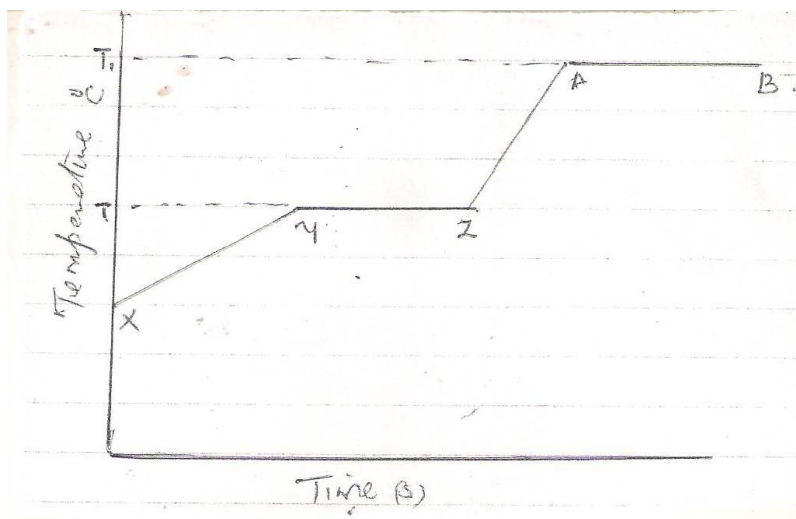
5.a. State two ways for determining the purity of substance. (2mks)

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b. The diagram below represents heat curves of a pure surface of solid. Study it and answer the questions that follows.



i. Region YZ and AB have one thing in common. State it and explain. (2mks)

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ii.State the physical state of substance at the following region

XY. (2mks)

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ZA (1mk)

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6.Define the following terms.

a.Solution 1mks

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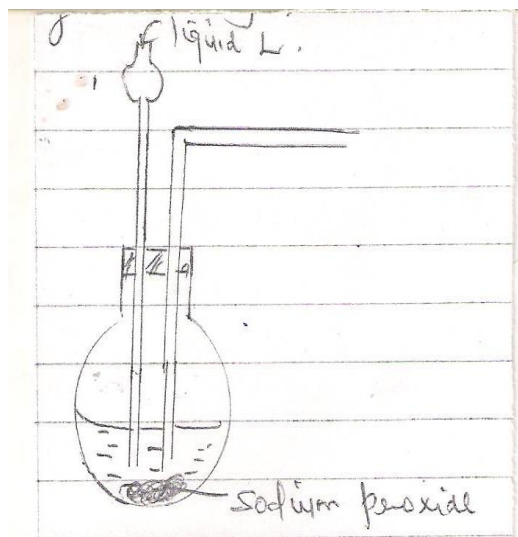
b.Saturated solution 1mks

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7.The following is a set up to prepare oxygen gas .Study it and answer the questions that follows.



a. Identify the mistake made when setting up the apparatus. (2mks)

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b. If the mistake was corrected, complete the diagram showing how to collect dry oxygen gas. (3mks)

c. Identify liquid L. (1mk)

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d. Write word equation for reaction at round-bottomed flask. (2mks)

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e. State two physical properties of oxygen. (2 mks)

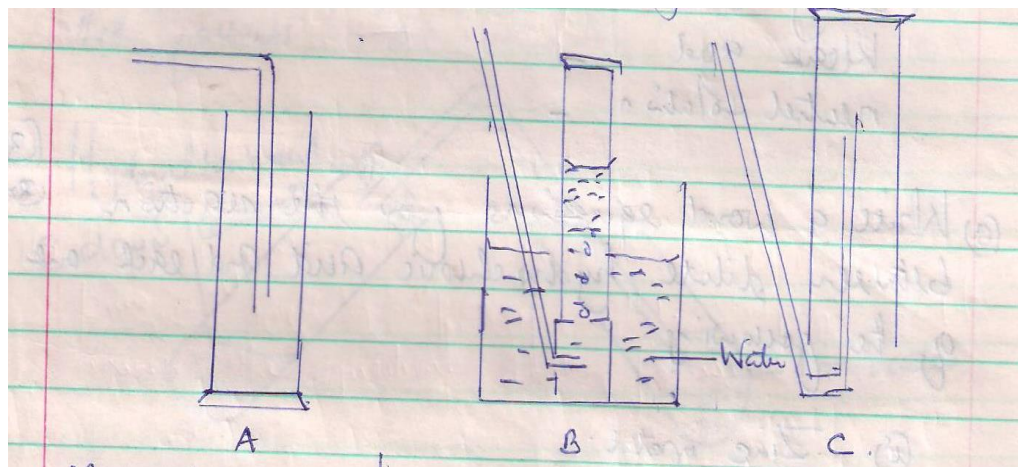
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8. Identify the following ways of collecting gases and state the reason.



9.a. State one advantage of the universal indicator over acid- base indicator. (1mk)

b. The pH of the solutions are given as below

Solution	pH
A	7
B	6
C	2
D	12
E	8

Which of the pH listed above is (5mks)

i. Strong acid -

ii. Weak base –

iii. Strong base-

iv. Weak acid-

v. Neutral solution-

c. Write word equations for the reaction between dilute hydrochloric acid and each one of the following

i. Zinc metal (2mks)

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ii. Calcium carbonate (2mks)

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iii. Magnesium Oxide (2mks)

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IV. Sodium hydroxide (2mks)

d. Miriam a form two student at Mutuma Trinity School was stung by wasp. Kanyotu advised her to apply sodium hydrogen carbonate. Explain. (2mks)

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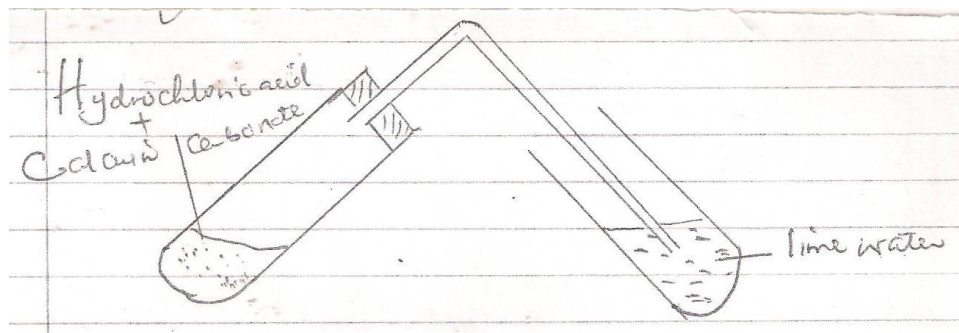
e. Why is not advisable not to use sodium hydroxide. (1mk)

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f. Study the set-up below and answer the questions that follows.



i. What are the products of the reaction between calcium carbonate and hydrochloric acid (give a word equation for the reaction). **(2mks)**

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ii. What would expect to observe in the limewater? **(1mk)**

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g.i) State two uses of bases. **(2mk)**

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ii) Two uses of acids. **(2mks)**

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 10. A student found a colourless liquid at laboratory. Describe two reagents he/she can use to determine the colourless liquid is water. (2mks)

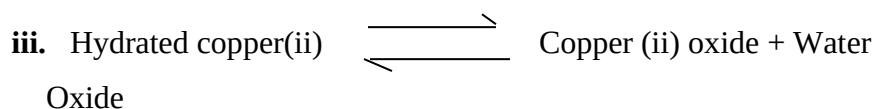
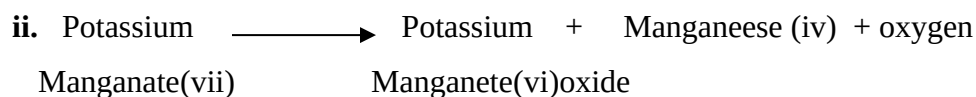
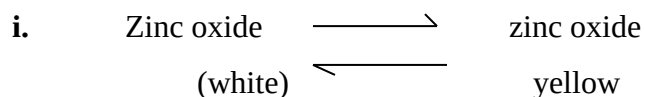
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 11. a.State two differences between permanent change and temporary physical changes. (4mks)

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 b.Study the following chemical equations (2mks)

heat



Identify the changes in (3mks)

Reaction I

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Reaction II

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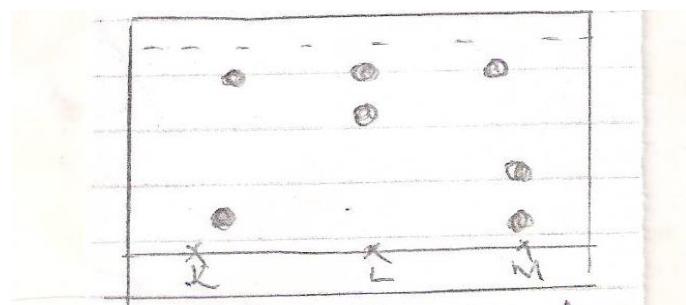
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Reaction III

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12. The diagram below represents a paper chromatogram for three brands of juices suspected to contain banned food colourings.



The results showed the presence of banned food colourings in L and M. One the same diagram.

a. Give the spots which show the banned food colourings. (2mks)

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b. Show solvent front. (1mk)

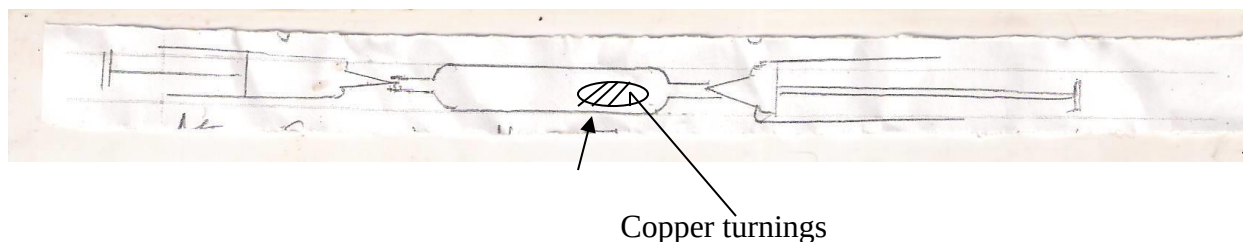
c. State two applications of chromatography. (2mks)

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13. The apparatus below were used to determine the volume of oxygen in air. About 200cm^3 of air were passed repeatedly and slowly from syringe A to syringe B, over heated copper turnings as shown in the diagram.



After sometime the volume of air syringe A was 160cm^3 and syringe B 0cm^3 .

- a. Calculate the percentage of oxygen in the initial sample of air. **(2mks)**
- b. Write down a word equation for the reaction that took place in the combustion tube. **(1mk)**

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- c. What are possible sources of error in the experiment. **(2mks)**

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- d. Explain why the air is passed slowly and repeatedly. **(1mk)**

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

SCHOOL:

ADM NO: **SIGN:** **DATE:**

FORM 1

CRE

END TERM 2

TIME: 2½ hours

INSTRUCTIONS:

Answer all the questions in the spaces provided.

1. Give four reasons why Christian religious education (C.R.E) is learnt in schools (4mks)

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2. a)Identify four poetic books in the old testament (4mks)

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b)Explain four reasons that make Christians not to be able to read the bible (4mks)

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3. Outline four responsibilities given to human beings during creation in the Old Testament (4mks)

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4. State four consequences of sin according to the book of Genesis (4mks)

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5.a) Outline six promises made to Abraham by God (6mks)

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b) State four ways in which Christians demonstrate their faith to God today (4mks)

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6.a) Explain three differences between the Jewish circumcision and traditional African circumcision

(6mks)

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b) Identify four forms of modern covenants in our society today

(4mks)

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7.a) Describe the call of Moses [exodus 3:1-22]

(8mks)

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b)Identify four attributes of God that a Christian can learn from the ten plagues (4mks)

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8a) Explain four ways in which God demonstrated his love to Israelites during the exodus. (8mks)

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b) In which ways do Christians go against the Decalogue today. (5mks)

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[illegible][illegible][illegible]

10. a) Explain four achievements of king David as king of Israel. (8mks)

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b) Outline the importance of the temple in Israel. (6mks)

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11a) State five reasons why Elijah faced danger in Israel. (5mks)

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BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

SCHOOL:

ADM NO: **SIGN:** **DATE:**

Instructions

Answer all the questions in this paper.
The question on functional writing should be answered on a separate foolscap.
This paper consists of the following sections.

- A. SECTION- A FUNCTIONAL WRITING (20mks)**
- B. CLOZE TEST (10mks)**
- C. ORAL SKILLS (15mks)**
- D. COMPREHENSION (20mks)**
- E. ORAL LITERATURE (20mks)**
- F. GRAMMAR (15mks)**

Total 100mks.

SECTION A FUNCTIONAL WRITING (20MKS)

Imagine you have presented a solo verse at the National Drama Festival and won. You have been invited for the award giving ceremony at State House Nairobi. Write a letter to your father informing him of the occasion and inviting him to attend the ceremony. Include any other relevant details.

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[illegible]

SECTION B. CLOZE TEST (10MKS)

It is exam time again and many parents and children Iover the country are experiencing pangs of tensions and nervousness. For some, exams are a matter of life and 2.....No wonder there are victims of 3.....fever, a miserable condition that causes stress and interferes 4.....the overall functioning of the body. Although stress in small doses 5.....necessary for motivation, it has been 6.....that the level of exam induced anxiety 7..... many parents and children is exceedingly high. As 8..... result, children's performance is partly 9.....by their ability to manage exam fever as 10.....to their level of intelligence.

SECTION C ORAL SKILLS

Provide another word that is pronounced exactly the same as the one given in each case.

- i. Hire
- ii. Weigh
- iii. Wholly
- iv. Blew
- v. Rain

(ii) Identify the odd word in the following sets of words. (4mks)

- (a) Thing thine although forth.
- (b) Enough fish emphasize shepherd.
- (c) Chore chit arch char
- (d) Jet genie game orgy.

(iii) Identify the silent letters in the following words. (3mks)

-whether

.....

Cupboard

.....

Sigh

.....

(iv) Give the words that are similar in meaning with the following words. (3mks)

-Candy

-Arise

Vast

SECTION D COMPREHENSION 20MKS

READ THE PASSAGE BELOW AND ANSWER THE QUESTIONS THAT FOLLOW.

WASTE DISPOSAL

Population in most cities is increasing at a very high rate. The high population results in high levels of activities and consumption. People in cities consume a lot of food and they buy groceries from shops which are wrapped in polythene bags. As a result, a lot of garbage is generated and disposed off within and outside the urban areas. The higher the population, the higher the amount of garbage generated. This means that there should be an efficient system of collecting and disposing off the garbage.

The effectiveness of waste management by city councils varies from country to country. In most African cities. A large amount of waste is not deposited in rubbish pits. It is dumped in the open. According to United Nations Environmental Programme (UNEP) statistics, sixty five percent of garbage in Tripoli, for example, is dumped in the open while eighty percent of garbage in Ismailia (Egypt) is also dumped in the open.

With extreme warm temperature such garbage tends to decompose rapidly causing serious health risks besides being a nuisance. One common method of desposing off waste is open burning. This contributes to the problem of urban air pollution.

In Nairobi, for example, it is common to find informal open burning sites. Most residents of city estates say that they are forced to burn waste in the open because the city council is incapable of dealing with the increasing amount of garbage. When waste is not disposed off effectively, it affects human health and damages the environment. For example, it has led to problems such as pollution of water sources, offensive odour, and increase in disease vectors, pests and scavengers.

However, there are examples of domestic waste being disposed off hygienically or being recycled. For example, in rural Egypt, there are organizations that promote environmental awareness among women by providing them with plastic bags so that they can collect their garbage. On the other hand, the youth are given the task of taking the garbage to a designated collection point. This scheme can be imported to the urban areas with a lot of success.

Another possible solution could be the introduction of reusable shopping bags in place of the usual polythene bags. Supermarkets and other businesses can pack customers’ goods in bags that are reusable. This would drastically reduce the amount of domestic waste generated every day.

(a) What is the result of high population in most cities. **(2mks)**

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(b) In most African cities, where are large amounts of waste deposited? **(1mk)**

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(c) How much waste does Egypt dump in the open? **(1mk)**

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(d) What two problems result from dumping waste in the open? **(2mks)**

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(e) Why is open burning of waste common? (1mk)

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(f) In what two ways can waste be successfully managed according to the passage. (2mks)

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(g) In note form, briefly state the various forms of waste disposal discussed in the passage. (4mks)

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(h) Why is Egypt mentioned in the passage? (1mk)

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.....
(i) What are the writer's feelings towards the city council. (2mks)

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.....
(j) The high population results in high levels of activities and consumption.

Rewrite beginning: Such(1mk)

(k) Explain the meaning of the following words and phrases as used in the passage.(3mks)

(i) Offensive odour

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(ii) Disposed off

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(iii) Decompose

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SECTION E ORAL LITERATURE

Read the following narrative and answer the questions that follow.

Once upon a time, the Hare and Cock were great friends. They did everything together including living in one house. They would every morning leave for the forest and plains to look for food to feed their family members. And every evening they would arrive home, each with their spoil, if they had been lucky and at times with nothing at all. Whichever the case, each one of them would be exhausted at the end of the hunt. At home, their wives would make the meal and soon each one of them would be sound asleep.

However, on one night the hare woke up in the middle of the night so as to attend to the call of nature. When he looked at the Cock, he saw it standing on one leg, with his head invisible. This disturbed him a bit and for the next three nights, he woke up at the same hour to ascertain whether what he had seen was true or a hoax. But on each night he woke up, he found the cock on one foot and presumably without his head. He then decided, the next morning on their way to the forest to ask his friend what happened to him every night because he always appeared transformed into a new image.

“I usually cut off my head and leg before going to sleep.” Said the Cock. Can you make me sleep like you? Without your head and leg?” inquired the Hare.

“Sure, sure,” replied the excited Cock. It was therefore agreed that the Cock would cut the hares head and legs the following night.

All day long, the hare waited with anticipation for night to arrive so that he could be transformed into a new being like his friend, the Cock.

The cock to him, time was not moving. After hours of waiting, evening arrived finding the hare in a very jovial mood. He hurriedly ate the evening meal that Mrs. Hare served, so that he could experience sleeping without his leg and head. In the meantime, the Cock was sharpening a big matchet ready to chop off the hare's head and legs.

Within minutes all was over. Mrs. Hare tried to wake up her husband to no avail. There was blood all over the place and no heart beat. When she realized what had actually happened to her husband, she wailed in the darkness. On looking up at the Cock, she saw that he had raised his matchet to cut off her head and decided to run for her dear life never to be seen again.

The Cock was very happy and called Hen to chop the pieces which were cooked and the whole family happily feasted on for days.

And that is why the saying goes “curiosity killed the cat.”

QUESTIONS

(a) With reasons classify the above narrative **(2mks)**

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(b) Briefly describe the character of

(i) The cock **(2mks)**

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(ii) The hare

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(c) Identify any two stylistic devices and explain their effect on the narrative. **(6mks)**

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(d) (i) The community makes a living from a certain economic activity. State it and give a reason for your answer. **(1mk)**

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(ii) State one social activity evident in the narrative. Give a reason for your answer. **(1mk)**

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(e) Give the meanings of the following words.

Anticipation

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Transformed **(2mks)**

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(f) What lessons do you learn from the narrative. **(2mks)**

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(g) “Curiosity killed the cat” (2mks)

(i)What do you call the above?
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(ii)What are the functions of the same in the narrative (2mks)
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GRAMMAR (15MKS)

(a) Put a or an before the following words or phrases.(5mks)

- (i)University
(ii)One –sided affair
(iii)Honest deal
(iv)Historical survey
(v)Unicorn

(b) Supply the missing prepositions in each of the following sentences.

- 1.You will find our house the end of the street.
2.It is strictly forbidden to parkthe middle of the road.

(c) Indicate the correct Noun category for the following words.(3mks)

- (i) Pastor
(ii)Kenya
(iii)Toy

(d) Use the correct form of the pronoun in the following.(5mks)

1.We scored as many goals as(they, them)

2.Let you andtry what we can do.(I, me)

3.Nobody butwas present (he, him)

4.Whom can I trust, if not.....? (she, her)

5.We are not as poor asare(they , them)

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

SCHOOL:

ADM NO: SIGN: DATE:

GEOGRAPHY: 312

FORM ONE

2 HOURS

Instructions to students

- ✓ Write your name, admission number and class in the spaces provided above.
- ✓ Attempt all questions.
- ✓ All your answers must be written in the spaces provided below each question.
- ✓ All answers must be written in English.

1.What is Geography? (2mks)

2.State four reasons why it is important to study Geography (4mks)

3.a) Name two branches studied in Geography (2mks)

b) Name three areas studied in practical Geography (3mks)

c) State the relationship between Geography and Mathematics (2mks)

4.a) What is the solar system? (2mks)

b) Name four components of the solar system (4mks)

c) State three characteristics of the earth (3mks)

5.a) Name two theories that explain the origin of the earth and the solar system (2mks)

b) Give three weaknesses of the passing star theory (3mks)

6.a) Describe the shape of the earth (2mks)

- b) Name the forces that resulted to the spherical shape of the earth **(3mks)**
- c) State four proofs that explain the spherical shape of the earth **(4mks)**
- d) Use a well labeled diagram to describe the lunar eclipse **(7mks)**

7.a) Name two movements of the earth (2mks)

- b) State four effects of earth's rotation **(4mks)**
- c) Differentiate summer solstice and winter solstice **(4mks)**
- d) What is Equinox? **(2mks)**

8.a) State three characteristics of the crust (3mks)

- b) Name three components of the atmosphere **(3mks)**
- c) Name three layers of the atmosphere **(3mks)**
- d) State three characteristics of the Troposphere **(3mks)**
- e) Differentiate positive lapse rate and negative lapse rate **(4mks)**

9.a) Define these terms (4mks)

- i) Statistics
- ii) Data

b) Name two types of statistical data (2mks)

c) State three sources of primary data (3mks)

d) State three sources of secondary data (3mks)

e) State three advantages of using observation as a method of data collection (3mks)

10. a) Differentiate between weather and climate (4mks)

- b) State three characteristics of a Stevenson's screen **(3mks)**
- c) State three reasons why data can be inaccurate **(3mks)**
- d) Explain briefly how you can measure rainfall using a rain gauge **(4mks)**

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

SCHOOL:

ADM NO: **SIGN:** **DATE:**

FORM 1

HISTORY AND GOVERNMENT

Answer all the questions in the spaces provided. **SECTION A**

- 1.State three characteristics of government. **(3mks)**
- 2.Identify two main periods of History. **2mks**
- 3.State why archaeology is an expensive source of information of history and government. **(2mks)**
- 4.State three aspects of oral traditions.**3mks**
- 5.Identify two ways through which archeologist obtain information on the history of Kenya. **(2mks)**
- 6.Identify two ways in which early man used stone tools. **(2mks)**
- 7.In what two ways did the Kenyan communities interact during the pre-colonial period. **(2mks)**
- 8.Identify two similarities between early agriculture in Mesopotamia and Egypt. **(2mks)**
- 9.Name the pre- historic sites in Kenya. **(2mks)**
10. Identify the main economic activity of the plain Nilotes during pre-colonial period. **(1mk)**
11. Give two characteristics of Homo Sapiens. **(2mks)**
12. State two ways in which the development of the upright posture improved the early man's way of life. **(2mks)**

Section B

13. a.State four forms of government. **4mks**
b.Identify five methods of dating. **(5mks)**
c.Outline the disadvantages of written sources of information. **(6mks)**
14. a.Discuss the main theories that explain the origin of man. **(6mks)**
b.State four species of Austroplithecus. **(4mks)**
c.What were the uses of fire during the middle Stone Age period? **(5mks)**

15. a. State five causes of food shortages in Africa today. **(5mks)**
b. Explain the characteristics of farming methods in Britain before the Agrarian Revolution. **(10mks)**
16. a. Why did the highland nilotes migrate from their original homeland during the pre-colonial period? **(5MKS)**
b. Explain five results of the migration and settlement of Bantus in Kenya. **(10mks)**
17. a. State five duties of council of elders among the Miji Kenda. **(5mks)**
b. Why was the office of the Oloibon important among the Maasai? **(5mks)**
c. Discuss the economic organization of the Borana in the 19th c. **(5mks)**

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

JINA:

SHULE:

NAMBARI YA USAJILI: SAHIHI: DATE:

KISWAHILI

MUHULA WA PILI

KIDATO CHA KWANZA

UFAHAMU (ALAMA 15)

Soma kifungu kifuatacho kasha ujibu maswali yanayofuata

Madhara ya sigara

Chimbuko la kuonekana kwa madhara ya tumbaku ilikuwa mwaka 1598 A.D. Madhara yake yaligundulika wakati ambapo makala ya kwanza ya kiafya iliandikwa nchini Uingereza ikieleza madhara ya sigara. Baada ya hapo zilifuatia tafiti mbalimbali zilizoendelea kugundua na kuelezea madhara yatokanayo na matumizi ya bidhaa za tumbaku kwa binadamu. Matokeo ya tafiti hizo yaliwezesha baadhi ya nchi kama vile Denmark, Uholanzi na Sweden kutunga sheria zinazozuia matumizi ya sigara. Katika sura ya pili, baadhi ya serikali zilikwazwa kiuchumi kwa kukosa pato la kodi zitokanazo na sigara na bidhaa zingine za tumbaku.

Mashirika na kampuni zinazojishughulisha na uzalishaji na uuzaji wa sigara hutenga fungu kubwa la fedha kwa ajili ya kutangaza bidhaa hiyo hatari kwa afya ya mwanadamu. Kila mwaka, Shirika la Afya Ulimwenguni hutenga siku maalumu ya kuelimisha jamii kuhusu athari za kiafya na madhara yatokanayo na matumizi ya tumbaku. **Maadhimisho** hayo hufanyika kila Mei 30 ya mwaka na kauli mbiu ya mwaka huu ilikuwa “Mkinge mwanamke kuwa mteja au kutumiwa kama mhamasishaji wa bidhaa zingine za tumbaku”.

Takwimu za shirika hilo zinabainisha kuwa asilimia 90 ya wavutaji sigara wameanzia umri wa miaka 18. Pia zimebainisha kuwa, kwa kila sekunde ipitayo, wastani wa mtu mmoja anafariki kutokana na madhara ya

tumbaku. Inakadiriwa kuwa ifikapo mwaka 2025, vifo vitaongezeka kwa asilimia 70. Utafiti umebainisha ifikapo mwaka 2030, watu milioni 10 watakuwa wamefariki kutokana na kuvuta sigara.

Uvutaji sigara unasababisha magonjwa mengi kama vile saratani ya mapafu, wendawazimu, **kupooza**, matatizo ya mfumo wa hewa, kutoboka utumbo, kunyonyoka nywele, kupungua kwa nguvu za kiume, maradhi ya ngozi na kadhalika. Madhara ya tumbaku hayajitokezi mara moja. Huchukua kipindi kirefu hata miaka 30, hali inayowafanya wengi kufumbia macho tahadhari iliyopo, wakiendelea kuteketezwa na uvutaji sigara. Anayevuta sigara huathirika mapafu kutokana na moshi na tindikali – aina ya kaboni iliyomo kwenye tumbaku inayomzuia kupumua na mapafu yake huanza kutunga usaha. Hali hiyo huchangia mtu kupata athari nyinginezo kama vile kichomi, kifua kikuu na saratani. Chembechembe ya nikotini iliyomo katika tumbaku ni hatari kwa mvutaji kwani humpotezea mtu hamu ya kula na anapata maumivu makali ya tumbo.

Wakati wote, mvutaji sigara hutoa harufu mbaya mdomoni, hali ambayo haipendezi katika maisha ya mwanadamu, na pia, huwa anawadhuru wengine. Mtu huyo anapopumua huchafua mazingira kwa kupumua hewa chafu na mbaya zaidi ambayo huwadhuru watu walio karibu naye kama vile mwenza katika ndoa (mke au mume), watoto au marafiki.

Inaelezwa kuwa sigara huathiri akili ya mwanadamu kutokana na kuwepo kwa chembechembe za ulevi. Hali hiyo ipo zaidi mtu anapotumia sigara kwa mara ya kwanza kabisa. Utafiti wa kitaalamu umethibitisha watoto wanaolelewa katika mazingira ya wazazi wanaotumia sigara kuathirika mishipa inayosafirisha damu kwenda katika moyo. Hali hiyo husababisha kuharibika kwa mfumo wa usafirishaji damu mwilini kwa watoto hao na mwishowe wanakuwa wahanga wa maradhi tofautitofauti.

Katika sura ya pili ya kisaikolojia, watoto wanaokaa na wazazi wanaovuta sigara huishia kuifuata tabia hiyo na huo ndio huwa mwanzo wa kudhoofika kwa maadili. Kwa wajawazito, sigara ina madhara kama vile kuathiri ukuaji wa mimba na kondo la nyuma au *placenta* kwa Kiingereza na kushindwa kusafirisha chakula vizuri kutoka kwa mama kwenda kwa moto. Madhara mengine ni moto kuzaliwa na uzito pungufu, hivyo kuathiri ukuaji wake. Kwa hivyo, ni vyema kufahamu kuwa ingawa sigara ni bidhaa inayotumiwa kama uraibu wakati wa kuivuta, madhara yake ni janga kubwa linalopaswa kupigwa vita bila kuchoka.

Maswali

(a) Madhara ya sigara yalionekana mwaka gani ?

(alama 1)

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(b) Ni nchi gani zilizokuwa za kwanza kubuni sheria mpya kuhusu uvutaji wa sigara ? (alama 1)

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(c) Uvutaji wa sigara una athari gani kwa anayevuta ? (alama 5)

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(d) Watoto wanaathiriwa vipi na sigara ? (alama 4)

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(e) Thibitisha kwamba uvutaji sigara hugharimu pesa nyingi. (alama 2)

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(f) Eleza maana ya msamiati ufuatayo kama ulivyotumiwa katika kifungu. (alama 2)

(i) Kupooza

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(ii) Maadhimisho

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LUGHA - SEHEMU B

(a) Kwa kuzingatia jinsi ya kutamka sauti za kswahili ni nini tofauti kati ya (alama 2)

(i) Irabu

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.....

(ii) Konsonanti

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.....

(b) Eleza maana ya dhana hizi (alama 2)

(i) Kiimbo

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(ii) Shadda

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(c) Taja vipashio vine vya lugha (alama 4)

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(d) Andika sentensi zifuatazo katika hali timilifu (alama 2)

(i) Mabati yananunuliwa

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(ii) Baba atakwenda dukani

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(e) Kanusha sentensi zifuatazo (alama 3)

(i) Nchi za Afrika mashariki zalemewa na madeni.

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(ii) Mbu amemuuma mtoto.

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(iii) Baba ameandika barua ndefu.

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(f) Zigeuze sentensi zifuatazo ziwe katika wakati uliopita (alama 2)

(i) Unapendelea kula nyama siku hizi.

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(ii) Opiyo alipenda kujiingiza katikati kama mchuzi wa ugali.

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(g) Andika kwa wingi (alama 2)

(i) Maziwa ya mtoto yalimwagwa na paka yule.

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(ii)Uvivu wa Juma unakera.

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(h) Akifisha sentensi zifuatazo (alama 2)

(i)nenda sokoni ukaninunulie sukari maziwa na mkate

.....
.....

(ii)Maadam umeshafika tuanze mkutano

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(i) Kamilisha jedwali lifuatalo kwa kuandika viambishi mwafaka kulingana na kitenzi (alama 5)

Kitenzi	Nafsi	Wakati	Kirejeshi	Kitendwa	Mzizi	kiishio
Aliyelima						
Aliyempiga						
Anaoupanda						
Alipika						
Nitavileta						

(j) Taja vipera vya fasihi simulizi (alama 2)

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.....

(k) Onyesha vielezi katika sentensi zifuatazo (alama 2)

Mama amepika chai tamu sana

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.....

Nitakutembelea kesho jioni

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.....

(l) Nyambua vitenzi vifuatavyo katika kauli ya kutendwa (alama 4)

Piga -.....

Jenga -.....

Fagia -.....

Chora -.....

(m) Taja na ueleze dhima ya lugha (alama 2)

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.....

(n) Andika sentensi zifuatazo katika hali ya mazoea (alama 2)

(i) Mchoraji yule anachora picha nzuri

.....
.....

(ii) Mpishi atapika chakula kitamu

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.....

(o) Bainisha maneno katika sentensi zifuatazo. (alama 4)

Salaala ! Majangili wamewaua ndovu kumi leo asubuhi

Wale watafungwa

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FASIHI (ALAMA 15)

Taja majukumu matano ya fasihi ya kiswahili (alama 5)

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Taja tofauti ya fasihi simulizi na fasihi andishi (alama 10)

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ISIMU JAMII (ALAMA 10)

Nunua sabuni mpya ya GRESHA. *It is new !* Harufu yake ni *poa*, inadumu siku yote, inalainisha ngozi. Ng’arisha nguo zako na sabuni ya GRESHA. GRESHA sabuni *poa !* Bei yake ni nafuu; shilingi 20/= tu. Kila mtu ainunua.

Maswali

(a) Hii ni lugha gani ? **(alama 2)**

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(b) Eleza sifa za lugha hii. **(alama 8)**

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Insha (Aalama 20)

Andika insha itakayomalizikia kwa maneno haya “Nilipopokea mezani chakula kitamu nilikumbuka maisha magumu niliyoyapitia, nikatambua kuwa si rahisi kupata chakula kama hicho bila juhudi maishani.”

[illegible]

[illegible]

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

SCHOOL:

ADM NO: **SIGN:** **DATE:**

MATHEMATICS FORM 1

END OF TERM II

TIME: 2 ½ HRS

INSTRUCTIONS

- 1.This paper consists of two sections I and II. **ANSWER ALL** Questions
- 2.All Answers and working **MUST** be written in the space provided.
- 3.Show all the necessary steps in your calculations.
- 4.Calculators **SHOULD NOT** be used.

Mathematical tables may be used.

SCORE

SECTION I	SECTION II	TOTAL SCORE

SECTION I (50 MARKS)

1. Evaluate : $\frac{-12 \div (-3) \times 4 - (-20)}{-6 \times 6 \div 3 + (-6)}$ (3mks)

2. factorize each of the following expression

a) $3px - 4ap + 3qx - qy$ (2mks)

b) $a^2 - 4ap - 4p + a$ (2mks)

3. The G.C.D of three numbers is 30 and their L.C.M is 900. Two of the numbers are 60 and 150. What is the least possible value of the third number? (3mks)

4. Three people Odawa, Mliwa and Amina contributed money to purchase a flour mill. Odawa contributed $\frac{1}{3}$ of the total amount, Mliwa contributed $\frac{3}{8}$ of the remaining amount and Amina contributed the rest of the money. The difference in contribution between Mliwa and Amina was sh.40,000. Calculate the price of the flour mill.
(4mks)

5. Arrange the following fractions in ascending order (3mks)
 $\frac{7}{8}$, $\frac{5}{6}$, $\frac{7}{12}$, $\frac{2}{3}$

6. Evaluate the following (3mks)
 $2.4 \times 3.2 \times 0.0096$
 $0.018 \times 0.064 \times 96$

7. In fourteen years time a mother will be twice as old as her son, four years ago the sum of their ages was 30 years. Find how old the mother was when the son was born. **(4mks)**

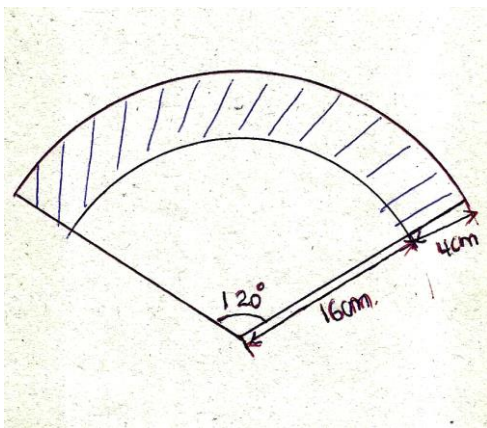
8. Three bells ring at intervals of 20 minutes, 30 minutes and 45 minutes. The bells will ring together at 12.40 pm. Find the time the bells had last rang together. **(3mks)**

9. When 10 is added to twice a number and the result doubled, the final result is 32. Find the number. **(3mks)**

10. Express $3.\overset{\bullet\bullet}{256}$ as a fraction **(3mks)**

11. Use mathematical tables to find the squareroot of 0.001952 (3mks)

12. The shaded region below shows the area swept out on a flat windscreen by a wiper, calculate the area of the region. (3mks)



13. The length of an arc of a circle is one-eighth the circumference of the circle. Find the angle subtended by this arc at the centre if the radius of the circle is 14cm . (Take $\pi = \frac{22}{7}$) (4mks)

14. The price of an article was raised by 20% and a week later the price was lowered by 20%. What is the new price if the original price was Sh.50. **(3mks)**

15. The acceleration due to gravity on the moon is 1.6N/Kg . An astronaut weighs 670N on the earth surface. Find

a) His mass **(2mks)**

b) His weight on the moon's surface **(2mks)**

SECTION II (50MKS)

16. Three business partners Bela, Joan and Trinity contributed Ksh.112,000, Ksh.128,000 and Ksh.128,000 respectively to start a business. They agreed to share their profit as follows:

30% to be shared equally

30% to be shared in the ratio of their contributions

40% to be retained for the running of the business.

If at the end of the year the business realized a profit of ksh. 1,350,000

Calculate :

a) The amount of money retained for the running of the business at the end of the year **(1mk)**

b) The difference between the amounts received by Trinity and Bela **(6mks)**

c) Express Joan's share as a percentage of the total amount of money shared between the three partners.
(3mks)

17. The net of the solid of a cuboid consists of a series of rectangles. By drawing the net of the solid determine:

- i. The surface area of the cuboid
- ii. Its volume

Given that it measures 6cm by 3cm by 2cm.

(10mks)

18. A tin has a square base whose sides measures 32cm. the vertical sides are 50cm high. Cake mixture is poured into the tin and leveled off at the 38cm mark above the base.

Find:

i. Surface area of the tin **(3mks)**

ii. The capacity of the tin **(3mks)**

iii. The volume of the cake mixture in the tin in cm^3 **(2mks)**

iv. The volume of the space unoccupied by the cake in cm^3 **(2mks)**

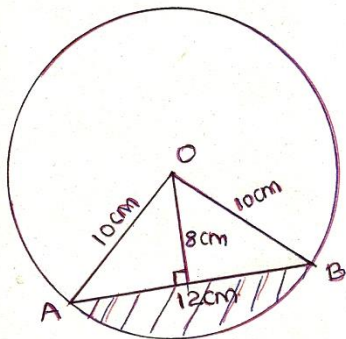
19. A cylindrical water tank with no top was constructed. If the diameter of the tank was 2.8m and height 4.8m. calculate:

a) Its volume (3mks)

b) Given that it was filled with a liquid of density 0.8g/cm^3 , calculate the mass of the liquid inside the cylinder (3mks)

c) Given that the cost of painting was 50 shillings for every 1m^2 , calculate the cost of painting the water tank (4mks)

20. Study the diagram below and answer the questions that follow.



Given that $\angle AOB = 120^\circ$, calculate

i. Area of the minor sector (3mks)

ii. Area of the triangle OAB (2mks)

iii. Area of the shaded region (2mks)

iv. By calculating the area of the circle first, calculate the area of unshaded region (3mks)

BRILLIANT STUDENTS FORM 1 END TERM 2 SERIES 3 EXAMS

NAME:

SCHOOL:

ADM NO: **SIGN:** **DATE:**

FORM 1 PHYSICS

END OF TERM 2

1.Name three physical quantities and their SI units. (6mks)

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2.Name the instruments you would use to measure each of the following:

(a) The length of a football field. (1mk)

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(b) The height of a 20 litre jerrican (1mk)

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(c) The circumference of your waist. (1mk)

3.A cylinder has a diameter of 4.2cm. How many times would a thread of 132cm be wound around the cylinder ? (3mks)

4. A page of a book measures 14.5cm x 21.4 cm. What is its area in square millimetres ? (3mks)

5. Define density and state its SI units. (2mks)

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6. The water level in a burette is 30cm³. If 55 drops of water fall from the burette and the average volume of one drop is 0.12cm³, what is the final water level in the burette ? (3mks)

7. A cube of iron of side 4cm has a mass of 512g. Find:

(a) The volume of the cube in m³ (3mks)

(b) The density of iron in kgm^{-3} (3mks)

8.What is pressure ? State its SI unit. (2mks)

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9.Explain the action of drinking straw. (3mks)

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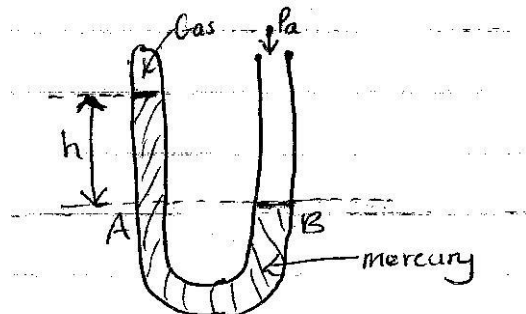
10. Using the crashing can experiment, explain using a diagram the existence of atmospheric pressure (5mks)

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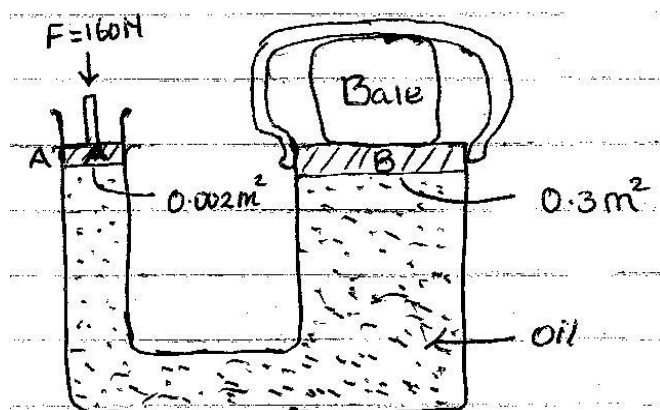
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11. The diagram below shows a mercury manometer. Some dry gas is present in the closed space in limb A, while limb B is open. If atmospheric pressure $P_a = 103\,000\text{ pa}$, $h = 30\text{mm}$ and density of mercury is $13\,600\text{kgm}^{-3}$. Determine pressure P_g of the gas. (Take $g = 10\text{Nkg}^{-1}$) (4mks)



12. The figure below shows a simple hydraulic press used to compress a bale. The cross-section areas of A and B are 0.002m^2 and 0.30m^2 respectively:



(a) Pressure exerted on the oil by the force applied at A. **(3mks)**

(b) Pressure exerted on B by the oil. **(2mks)**

(c) Force produced on B compressing the bale. **(3mks)**

13. A sea diver is 35m below the surface of sea water. If the density of the sea water is 1.03gcm^{-3} and g is 10N/kg . Determine the total pressure on him. (Atmospheric pressure = $103\,000\text{Nm}^{-2}$) **(3mks)**

- 14.** A brick 20cm long, 10cm wide and 5cm thick has a mass of 500g. Determine the:
- (a)** Greatest pressure that can be exerted by the brick on a flat surface. **(3mks)**
- (b)** Least pressure that can be exerted by the brick on a flat surface. (Take $g = 10\text{N/kg}$) **(3mks)**
- 15.** Define matter and give the three states of matter. **(4mks)**
- 16.** Explain why the density of a gas is much less than that of a solid or liquid. **(2mks)**

17. Explain the following:

(a) It is possible to compress gases but not solids and liquids. (2mks)

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(b) A perfume sprayed at one corner of a room spreads quickly to the entire room. (2mks).

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18. A smoke cell contains a mixture of trapped air and smoke. The cell is brightly lit and viewed through a microscope. Small bright specks are seen dancing in a random manner.

(a) What are these small bright sperks ? (1mk)

(b) Why do they move in a manner described above ? (2mks)

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19. Using pollen grains placed in water, explain how their motion supports the idea that matter is not continuous. (2mks)

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20. Define force and give its SI units. (2mks)

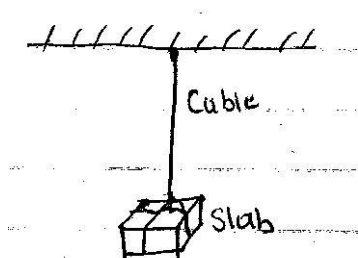
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21. A concrete slab of mass 90kg is held by a steel cable of a crane as shown in the fig below.



(a) Draw and name the forces acting on the slab. (2mks)

(b) Determine the tension in the cable. (3mks)

22. When water is poured on a dry glass slab it spreads uniformly but it forms spherical droplets on a waxed glass slab. Explain. **(4mks)**

23. A man has a mass of 70kg. Determine

(a) His weight on earth, where the gravitational field strength is 10N/kg. **(3mks)**

(b) His weight on the moon, where the gravitational field strength is 1.7N/kg. **(3mks)**

24. A mass of 7.5kg has weight of 30N on a certain planet. Calculate the acceleration due to gravity on this planet. **(3mks)**

25. A spring stretches by 6cm when supporting a load of 15N.

(a) How much would it stretch when supporting a load of 5kg? **(3mks)**

(b) What load would make the spring extend by 25mm ? **(3mks)**

26. Describe a method that can be used to open a tight lid of a bottle without damaging it. (2mks).

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